Neurocognitive and Evolutionary Architecture of Visual Perceptual Completion: Mechanisms of Sensory Interpolation and Object Synthesis

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Abstract:

The human visual system constantly encounters fragmented sensory inputs due to environmental occlusions and anatomical deficits like the blind spot. Despite this, perception remains phenomenologically continuous. This paper comprehensively reviews the neurocognitive, developmental, and evolutionary architectures underlying visual perceptual completion—the brain's capacity to synthesize complete object representations from partial ("half-seen") stimuli. We trace the historical foundations of Gestalt psychology to modern neurophysiological discoveries, distinguishing between modal and amodal completion mechanisms.[1] Evidence demonstrates that while early visual cortices (V1/V2) utilize structurally driven isomorphic interpolation, the lateral occipital complex (LOC) and inferotemporal (IT) cortex govern knowledge-driven, symbolic completion.[2] Furthermore, this review examines the extreme manifestation of top-down priors through "one-shot perceptual learning," detailing recent ultra-high-field fMRI and intracranial EEG findings that localize these instantaneous priors to the high-level visual cortex. Framed within the Bayesian predictive coding hypothesis and evolutionary error management theory, we argue that visual completion is not merely a neural error, but a critical, metabolically expensive adaptation driven by the ecological arms race of predator-prey dynamics and camouflage.

Keywords: *Amodal Completion, Modal Completion, One-Shot Perceptual Learning, Predictive Coding, Gestalt Psychology, Visual Cortex, Evolutionary Ecology*

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1. Introduction to the Computational Challenge of Vision

The human visual system is continuously confronted with an immense and inescapable computational challenge: the retinal projection of the external world is inherently fragmented, two-dimensional, and riddled with profound spatial and temporal ambiguities. Objects in natural, ecologically valid environments rarely present themselves in full, unobstructed isolation. Instead, they are frequently obstructed by other elements, cast in uneven and deceptive illumination, or partially obscured by atmospheric conditions. Furthermore, the physiological and anatomical architecture of the eye itself introduces permanent spatial voids in the visual field.[3] The most prominent of these is the optic disc, the locus where the optic nerve exits the retina, which entirely lacks photoreceptors and creates a permanent functional scotoma known as the blind spot.[3]

Despite these profound deficits in direct sensory stimulation, human conscious perception rarely registers a fragmented or perforated world. Instead, the visual environment is experienced as a seamless, continuous, and highly organized three-dimensional spatial reality.[2] The brain achieves this seamless phenomenological experience through an extraordinarily sophisticated, multi-tiered set of neural operations collectively termed "perceptual completion" or "filling-in".[2] Perceptual completion refers to the visual system's remarkable capacity to interpolate missing information, synthesize unperceived features, and construct whole representations from partial, disjointed sensory inputs.[4] Essentially, when the human eye perceives only "half" of an object, the brain intrinsically and automatically creates the other half, ensuring that organisms can interact with their environments based on the genuine physical properties of objects rather than the accidental and chaotic properties of their retinal images.[5]

This interpolative capability is generally categorized into two primary phenomenological and psychophysical distinct processes: amodal completion and modal completion.[1] Amodal completion occurs when portions of an object are visibly occluded by another opaque structure situated in the foreground.[5] For instance, when an animal, such as a cat, is standing behind a picket fence, the visual system successfully represents the cat as a continuous, unified entity extending securely behind the occluding pickets, even though the raw sensory data only provides disjointed vertical slices of the animal's geometry.[5] The completed portions are termed "amodal" because they are registered and represented by the cognitive system without the associated sensory phenomenology of brightness, contrast, or color in the occluded regions; the observer knows the object is there and perceives its volume, but does not literally see the sensory surface.[8]

Conversely, modal completion involves the perception of distinct edges, surfaces, and contours that possess vivid sensory qualities—such as measurable brightness or color boundaries—despite the absolute and verifiable absence of local luminance or color contrast in the physical stimulus.[10] The most famous demonstration of modal completion is the Kanizsa figure, wherein strategically aligned "pac-man" inducer shapes generate the vivid optical illusion of a bright, solid geometric form superimposed over the background.[11] In this scenario, the visual system hallucinate a subjective contour and a phenomenal surface that appears modally real to the observer, functioning as a proactive solution to the inverse optics problem.[13] Both amodal and modal completion are foundational to biological vision, enabling rapid object recognition, spatial navigation, and evolutionary survival.[10] The subsequent sections of this report will exhaustively detail the theoretical frameworks, developmental timelines, neurobiological underpinnings, computational predictive principles, and evolutionary ecology driving the brain's capacity to synthesize coherent wholes from fragmented visual parts.

2. Historical and Theoretical Frameworks: The Gestalt Paradigm

The scientific inquiry into the brain's interpolative functions has its deepest roots in the early twentieth century with the emergence of Gestalt psychology in Germany and Austria.[16] Founded by visionaries such as Max Wertheimer, Kurt Koffka, and Wolfgang Köhler, the Gestalt school developed as a direct, aggressive rejection of the elementalist and structuralist paradigms championed at the time by Wilhelm Wundt and Edward B. Titchener.[16] Structuralism posited that perception could be entirely reduced to its atomic sensory components—a summative assembly of individual lines, colors, and luminances that strictly mirrored the physical world.[16] Gestalt theorists, conversely, demonstrated through rigorous psychophysical experimentation that the phenomenological whole is fundamentally different from, and computationally precedes, the sum of its disparate parts.[11] The visual system does not merely add pieces together; it imposes an overarching architecture upon them.

The Gestalt school formalized the human brain's propensity to organize visual chaos through a comprehensive set of heuristic guidelines known as the Laws of Perceptual Organization.[16] These principles describe the automatic, stimulus-driven algorithms the visual system employs to group disparate fragments into cohesive objects, dictating the very mechanics of amodal and modal synthesis.[17]

2.1 Principles of Grouping and Object Formation

Several core Gestalt laws directly govern the mechanics of perceptual completion, operating across varying levels of spatial integration.

The Law of Closure describes the inherent psychological and neural inclination to seek completeness in the visual array.[12] When confronted with a visual pattern containing geometric gaps, the brain effortlessly and rapidly fills in the missing spatial information to perceive a closed, unified figure.[12] The Law of Closure dictates that enclosed, solid shapes are statistically more probable in natural ecological environments than random assortments of open curves, prompting the visual system to enforce closure as a stabilizing, probabilistic heuristic.[18]

The Law of Good Continuation, often modernized in computational vision science as the principle of "relatability," serves as the primary bottom-up geometric constraint for amodal contour interpolation.[5] When two disjointed contour fragments can be connected by a smooth, monotonic curve devoid of abrupt inflections or sharp angles, the visual system is highly likely to mathematically interpolate a connection between them.[5] Relatability requires that the visible edges of a partially occluded object align seamlessly, indicating that they belong to the same underlying physical boundary.[5]

The Law of Prägnanz, occasionally referred to as figural simplicity or the principle of good figure, dictates that the visual system will always default to the simplest, most regular, and most globally symmetric interpretation of an ambiguous or incomplete stimulus.[5] If a pattern of occlusion can be mathematically resolved into a highly complex, chaotic asymmetric shape or a simple, predictable symmetric shape, the global minimization algorithms of the brain will automatically synthesize the symmetric completion.[5] This global completion acts as a top-level organizational principle that can override local ambiguities.[20]

Finally, the Laws of Proximity and Similarity establish the foundational segments upon which closure and continuation subsequently operate.[5] Elements that are close to one another in physical spatial coordinates, or share specific feature domains such as color, surface texture, or orientation, are preferentially grouped together early in the visual processing stream.[5] These principles have profound, measurable impacts on human interaction with complex visual environments, including the built environment. For example, contemporary eye-tracking studies assessing architectural facades demonstrate that buildings designed utilizing Gestalt proximity principles receive significantly higher

aesthetic ratings, elicit shorter fixation durations, and demand less cognitive processing time than chaotic or overly complex facades.[21] Conversely, manipulations of similarity and complexity alter aesthetic preference, proving that these primitive grouping mechanisms deeply influence high-level evaluations of the environment.[21]

2.2 The Kanizsa Paradigm and Temporal Integration

The Kanizsa figures, particularly the Kanizsa triangle, stand as the quintessential psychophysical probes for understanding the Law of Closure and modal completion.[11] In a Kanizsa display, spatially isolated elements are configured such that their collinear edges strongly imply the presence of an occluding surface. The observer perceives an illusory contour—a sharp edge defining a shape that possesses higher apparent luminance than the background, despite total physical homogeneity in the luminance profile of the actual space.[11]

This illusion proves definitively that the visual system actively posits structural hypotheses. The perception of the Kanizsa figure is notably resistant to conscious cognitive override; even when an observer intellectually knows no triangle exists, the visual system enforces the modal percept.[11] Recent psychophysical evaluations involving rapid serial visual presentation tasks suggest that the temporal integration of stimuli is significantly enhanced when the constituent parts form a Kanizsa shape or exhibit common fate.[22] Temporal target integration produces a single, combined representation of visually compatible stimuli, facilitating rapid identification under high cognitive load.[22] The presence of a Kanizsa shape exerts a unifying effect on the perception of successive targets, indicating that the synthesis of the whole occurs at an extremely rapid, pre-attentive stage of visual processing designed to stabilize perception amidst rapid eye movements and fleeting exposures.[22]

3. Ontogeny and Developmental Trajectories of Perceptual Synthesis

A critical and ongoing question in developmental cognitive psychology is whether these Gestalt completion mechanisms represent innate, neurobiological hardwiring or whether they are gradually acquired through continuous sensorimotor interaction with the postnatal environment.[23] Robust empirical evidence indicates a protracted, highly specific, stage-like development of these capabilities throughout infancy and early childhood. While newborn visual systems are demonstrably capable of processing local features such as high-contrast edges and motion, the sophisticated capacity to bind these

features across occlusions and spatial voids requires significant cortical maturation.[24]

Studies utilizing preferential looking and habituation paradigms suggest that modal completion—the perception of illusory contours, as observed in Kanizsa figures—emerges earlier in the developmental timeline, typically robustly observable by 3 to 4 months of age.[7] In contrast, amodal completion—the cognitive representation of occluded objects residing in the background—appears to consolidate slightly later, firmly developing around 5 to 6 months of age.[7] This chronological disparity holds profound theoretical implications. It implies that synthesizing a foreground occluder from visible local spatial cues may impose lower computational and representational demands on the developing cortex than generating an abstract, non-sensory representation of a background object for which sensory data is fundamentally absent.[7]

Furthermore, the types of perceptual completion unfold in distinct domains. Developmental research distinguishes between spatial completion (the perception of partly hidden surfaces as continuous in two dimensions), spatiotemporal completion (the understanding that objects continue to exist upon becoming fully occluded over time), and 3D object completion (the perception of objects as solid volumes in three-dimensional space with unseen backs and sides).[24] The delayed onset of complex amodal completion aligns securely with the gradual maturation of long-range horizontal connections within the primary visual cortex and the delayed myelination of top-down feedback projections from the inferotemporal cortex.[24]

Even after these capabilities emerge in infancy, the binding processes involved in the extraction of illusory contours undergo a protracted period of refinement well into childhood.[25] Psychophysical assessments involving children aged 3 to 6 years reveal that their responsiveness to subjective contours—such as Kanizsa triangles and Ehrenstein figures (which utilize converging lines rather than pac-men to induce a central illusory surface)—is not entirely adult-like.[25] While six-month-old infants can distinguish between subjective and non-illusory displays, they frequently fail to detect the subjective figures in complex visual search tasks, indicating that while the primary sensory mechanisms are online, the attentional and integrational networks required to deploy these percepts in cluttered environments continue to mature.[25]

Perceptual Capability	Typical Onset Age	Mechanistic Description	Sensory Domain
Local Feature Detection	Birth	Identification of raw luminance, high-contrast edges, and motion.	Photoreceptors, Retinal Ganglion Cells.
Modal Completion	3 - 4 Months	Synthesis of subjective contours and illusory surfaces (e.g., Kanizsa figures).	V1/V2 recurrent lateral activity.
Amodal Completion	5 - 6 Months	Representation of occluded, continuous objects in the background.	LOC and IT cortex integration.
Complex Visual Search	3 - 6 Years	Deployment of contour synthesis within cluttered, noisy visual arrays.	Fronto-parietal attentional networks.

4. Neurophysiological Architecture of Visual Completion

The construction of a coherent visual whole from half-seen objects is not localized to a

single cortical hub; rather, it is distributed across an intricate, bidirectional hierarchy of regions within the visual cortex, stretching from the primary visual cortex (V1) at the occipital pole to the inferior temporal (IT) cortex.[1] Understanding this architecture requires mapping the transformations that visual information undergoes as it ascends and descends the cortical hierarchy. Visual processes often labeled as "intermediate level vision" involve sequences of mappings from one representation format to another.[26]

4.1 Early Visual Cortex: Spatiotemporal Dynamics in V1 and V2

The primary visual cortex (V1) acts as the entry point for cortical visual processing, characterized by neurons with highly localized receptive fields tuned to precise spatial locations, spatial frequencies, binocular disparities, and local contour orientations.[26] Historically, V1 was conceptualized merely as a passive feature-extraction map—a biological bitmap translating retinal activation.[26] However, modern neurophysiology comprehensively demonstrates that V1 plays an aggressively active role in visual completion. Electrophysiological and neuroimaging studies show that when an illusory contour (such as the edge of a Kanizsa triangle) crosses the receptive field of a V1 neuron, the neuron exhibits significantly increased firing rates, despite the absolute lack of sensory stimulation in that specific retinal location.[10]

Area V2, the secondary visual cortex, demonstrates even more robust and complex responses to illusory and occluded contours.[14] The receptive fields of V2 neurons are considerably larger than those in V1, enabling them to integrate structural information across wider spatial expanses.[26] Spiking rates in V2 correlate strongly with the perception of subjective contours and are acutely modulated by the orientation, distance, and geometric alignment of surrounding inducers.[14] The temporal dimension of this contour interpolation is remarkably fast but highly structured. Experimental data utilizing visual occlusion indicate that the amodal completion process in early visual cortex typically takes between 100 and 200 milliseconds post-stimulus onset.[19] Furthermore, this timing is subject to profound orientational anisotropies; the time required for the completion process depends directly on contour orientation, such that vertically and horizontally oriented contours are completed significantly faster than oblique contours, reflecting the statistical dominance of cardinal orientations in natural visual ecologies.[19]

Crucially, the interpolation mechanism in the early visual cortex is heavily reliant on the structural integrity of lateral horizontal connections cascading across the cortical surface.[31] A compelling demonstration of this involves the interhemispheric divide.[31] The left visual field is processed exclusively by the right hemisphere, and the right visual field

by the left. When the fragments of an occluded object or the inducers of an illusory contour span across the vertical meridian of the visual field, perceptual completion is significantly and measurably impaired compared to when the entire configuration resides entirely within a single visual hemifield (either left or right).[31] This interhemispheric deficit highlights a critical limitation in cross-hemispheric integration.[32] It robustly supports the existence of specialized completion mechanisms rooted in the early visual cortical areas (V1/V2), as these early areas are vastly more sensitive to the interhemispheric divide than higher-order areas, which possess bilateral receptive fields capable of bridging the meridian effortlessly.[31]

4.2 Higher-Level Visual Cortex: V4, IT, and the Lateral Occipital Complex (LOC)

As visual signals progress along the ventral stream from V2 to area V4, representations shift from localized line segments to intermediate shape features, complex patterns, and contour curvatures.[2] Neurons in V4 are sensitive to partial occlusions and dramatically modulate their responses based on the geometrical properties of the completed shape, suggesting V4 acts as a critical transitional layer where local Gestalt groupings are coalesced into regional boundaries.[2] Furthermore, the shape selectivity of individual V4 neurons is explicitly shown to be modulated by higher-level executive areas, specifically the ventrolateral prefrontal cortex, indicating that attention and cognitive state dynamically tune mid-level feature extraction.[1] Attention exerts a massive influence even on baseline neural activity; spontaneous firing rates in areas V2 and V4 are found to be 30–40% higher when attention is directed inside a neuron's receptive field compared to outside, even when no stimulus is present, reflecting a preparatory top-down signal that biases the network to connect anticipated fragments.[33]

The definitive assembly of the amodally completed object occurs in the Lateral Occipital Complex (LOC) and the broader inferotemporal (IT) cortex.[1] The LOC is located anterior to V4 and contains neurons with massive receptive fields capable of pooling information from vast portions of the visual field, rendering them highly responsive to complete, familiar objects rather than isolated fragments.[1] Functional Magnetic Resonance Imaging (fMRI) studies utilizing sophisticated repetition suppression paradigms have demonstrated definitively that the LOC represents amodally completed objects with the exact same neural signature as their fully visible, unoccluded counterparts.[2]

Recent repetition suppression paradigms have further disentangled the roles of structure (e.g., linear continuation) and knowledge (e.g., object familiarity) in this region. The

LOC exhibits significant BOLD signal suppression when a completed stimulus is compatible with either structural rules or semantic knowledge, and shows maximum prediction error when a shape violates both.[2] In contrast, the Early Visual Cortex (EVC) shows no such differential responses based on structural or knowledge-based expectations for amodal completion, cementing the LOC as the primary neural locus for complex, knowledge-integrated object continuity.[2] The fusiform face area (FFA) shows similar invariance, treating partially occluded faces identically to whole faces.[1]

5. The Isomorphic vs. Symbolic Representation Debate

A major theoretical and philosophical fissure in the neuroscience of perceptual completion involves the exact nature of the neural representation constructed by the brain: is the representation physically isomorphic, or purely symbolic?.[4]

The isomorphic hypothesis postulates that perceptual filling-in involves a literal, spatial spreading of neural activity across the topographically organized retinotopic maps of the early visual cortex.[4] In this view, if the brain needs to fill in a missing surface or contour, lateral horizontal connections physically propagate excitatory neural signals across the physical cortical tissue corresponding to the scotoma or occluded area, creating an active neural "bridge".[4] Evidence from high-resolution fMRI and single-unit electrophysiological mapping of V1 robustly supports this, showing increased BOLD signals and spiking rates in the exact retinotopic coordinates of the "hallucinated" missing regions, proving that the brain effectively paints the missing pixels into the primary visual array.[4]

Conversely, the symbolic (or cognitive) hypothesis, often championed by philosophers like Daniel Dennett, argues against the necessity of a physical neural bridge, criticizing it as a modern manifestation of "Cartesian materialism".[4] Critics suggest that filling-in might be achieved by higher-order areas simply applying a symbolic "tag" or cognitive pointer to a region, effectively declaring the belief that "this area is continuous with its surroundings" without needing to physically activate the spatial map in V1.[4] From this perspective, lower-level activations might merely be epiphenomenal feedback rather than the core mechanism of perception itself.[30] Dennett argued that the brain simply "ignores an absence" rather than generating a presence.[34]

However, the empirical evidence heavily contradicts the purely symbolic view. The extremely rapid timeline of amodal completion—occurring within 100-200 milliseconds

of retinal stimulation—poses a severe temporal challenge to purely symbolic, high-level accounts.[30] There is simply insufficient synaptic time for signals to ascend the entire cortical hierarchy, form a complex belief-based cognitive representation, and then trickle all the way back down to modulate sensory experience.[30] Furthermore, completion has measurable psychophysical effects (like lightness induction and adaptation) that strictly depend on neural signals representing a physical presence, rather than just ignoring an absence.[27] When surround luminance is modulated, activity in V1 correlates directly with the illusory change in the lightness of the perceived disk, proving early visual cortex is physically generating the filled-in experience.[27] The current scientific consensus leans toward a highly integrated hybrid model: localized isomorphic filling-in operates robustly in early cortices for bounded regions and modal contours, while large-scale amodal object continuity relies heavily on symbolic, object-level completion in the LOC.[3]

6. Predictive Processing and Bayesian Inference

The classical, twentieth-century view of visual processing treated the brain as a passive feedforward processor, where retinal inputs sequentially construct increasingly complex representations in a strictly bottom-up manner.[28] However, the complex phenomena of perceptual completion necessitate an active, inferential model of vision. The most robust and currently favored computational framework for understanding how the brain creates the missing half is the theory of Hierarchical Predictive Coding, grounded deeply in the Bayesian brain hypothesis.[3]

6.1 Generative Models and Error Minimization

Predictive coding postulates that the brain operates fundamentally as a Bayesian inference engine.[38] Instead of passively waiting for and processing sensory data, the cerebral cortex continuously generates internal "generative models" of the environment based on past experience and current context.[38] These complex mental models are used to simulate predicted sensory inputs, which are then cascaded backward (top-down) from higher cortical areas (such as the IT cortex and LOC) to lower sensory areas (such as V1 and V2).[38] The roots of this predictive framework trace back to Helmholtz's concept of "unconscious inference" and Jerome Bruner's "New Look" psychology, which posited that expectations fundamentally shape perception.[38]

When the predicted neural input perfectly matches the actual bottom-up sensory input arriving from the retina, the incoming sensory signal is successfully "explained away"

and neurologically suppressed.[29] This suppression maximizes metabolic efficiency, as the brain only expends energy processing what it did not expect. However, when there is a mismatch—for example, when an expected continuous object is suddenly interrupted by an occluder—a "prediction error" is generated.[38] This raw prediction error is then propagated forward up the cortical hierarchy to update and refine the internal models.[38]

Neural models for the implementation of predictive coding invoke specific functional classes of neurons. They require neurons for generating predictions, neurons for producing reciprocal error signals, and crucially, "precision" neurons.[39] Precision neurons are tasked with regulating the magnitude and weight of the error signals, entirely contingent upon the confidence placed upon the prediction—that is, the statistical reliability and behavioral utility of the sensory data in the current environmental context.[39]

6.2 Feedback Modulations in the Visual Cortex

Predictive coding provides an extraordinarily powerful explanation for why early visual cortices exhibit active responses to physically absent illusory shapes.[28] When a Kanizsa triangle is presented, the LOC instantly identifies the global geometric pattern and generates a prediction regarding the presence of a continuous, bright edge.[29] This high-level prediction is sent via recurrent feedback connections down to V1 and V2. Neurons in V1 whose receptive fields perfectly align with the expected illusory edge receive strong excitatory feedback, resulting in an enhanced BOLD response despite the lack of retinal input.[1]

Simultaneously, the neural representation of the Kanizsa "pac-men" inducers themselves—the actual, physical sensory data—is actively suppressed in early visual cortex.[29] Why? Because their presence has been perfectly predicted and accounted for by the higher-order generative model of the triangle.[29] This delicate, continuous interplay of feedforward error signals ascending the hierarchy and feedback predictive models descending it forms the dynamic, oscillating equilibrium of visual completion.[1]

This framework is also applied to understand the filling-in of the physiological blind spot. Computational models suggest that filling-in at the blind spot is a pure manifestation of hierarchical predictive coding of natural images.[3] The mechanism is driven by binocular neurons situated in the deep cortical layers (layers 5 and 6).[3] During early development, these neurons undergo "pre-training" via feedforward input from the non-blind-spot eye, allowing priors and predictions to co-evolve.[3] When input from one eye is missing due to

the blind spot, the predictive model seamlessly extrapolates the surrounding natural image statistics across the void via V2-V1 cortico-cortical interactions.[33]

Conversely, breakdowns in this predictive mechanism explain severe perceptual pathologies. In conditions like schizophrenia, the integration of feedback priors and feedforward sensory inputs is fundamentally disrupted.[28] Patients often exhibit altered susceptibilities to visual illusions because their predictive models fail to appropriately override or contextualize fragmented sensory data, highlighting how deeply dependent normal perception is on active, top-down neural hallucination.[28]

7. Epistemological Implications: Structure vs. Knowledge

The predictive framework naturally accommodates the fierce ongoing debate regarding the cognitive penetrability of perception—specifically, the degree to which amodal completion relies strictly on bottom-up "Structure" versus top-down "Knowledge".[2]

Structure-driven completion relies entirely on low-level, automatic factors, fundamentally grounded in the Gestalt principles of good continuation and local contour geometry.[2] It operates automatically to connect aligned edges irrespective of the object's identity. In contrast, Knowledge-driven completion relies heavily on high-level cognitive factors, specifically object familiarity, semantic expectations, and specific material properties.[2]

While early Gestaltists and researchers like Kanizsa argued that completion is purely structurally driven and utterly resistant to top-down knowledge (demonstrated by illusions that persist despite contrary intellectual knowledge), contemporary fMRI research reveals a vastly more complex, hybridized reality.[2] Investigations utilizing the aforementioned LOC repetition suppression paradigms reveal that both structural geometry and semantic knowledge significantly modulate neural responses.[2]

When a naturalistic object is occluded, the LOC evaluates the completion based on both how smoothly the edges connect (structure) and whether the resulting shape constitutes a familiar entity (knowledge).[2] If a proposed completion violates both structural geometry and semantic familiarity (e.g., resulting in a highly unnatural, jagged configuration of an otherwise recognizable fruit), the LOC exhibits minimal suppression, indicating high prediction error and neural deviance.[2] Explicit learning can also strongly influence the preferred completion for occlusion patterns, especially in ambiguous cases where local and global cues conflict.[2]

The profound implication of top-down knowledge deeply influencing amodal completion is strictly epistemological.[6] Knowledge is theoretically valuable because it objectively tracks truth.[6] Perception is supposed to be the ultimate, unbiased mechanism for acquiring that knowledge.[6] However, if our visual system completes objects based partly on our preexisting thoughts, beliefs, and learned expectations, then pure, unbiased sensory perception is an illusion.[6] Because almost all natural viewing involves profound occlusion, our most basic conscious acquisition of visual reality is perpetually colored by learned priors.[6] This introduces a form of vicious philosophical circularity: our beliefs about the world are justified by our perceptual states, but those very perceptual states are actively constructed by our pre-existing beliefs via top-down amodal completion.[6] If perception is continuously molded by what we already think we know, it cannot serve as a purely unbiased mechanism for learning about the objective world.[66] This "New Look" psychology paradigm dramatically challenges the modularity of mind, suggesting perception is deeply cognitively penetrable.[38]

8. One-Shot Perceptual Learning: The Extremes of Prior Influence

The absolute most dramatic manifestation of knowledge-driven perception and top-down influence is the phenomenon of "one-shot perceptual learning".[15] In standard machine learning algorithms and classical biological conditioning, acquiring a stable, reliable representation typically requires thousands of iterative, slow exposures.[42] However, the human brain possesses the astonishing capability to irreversibly alter its entire perceptual architecture after a single, momentary visual exposure.[15]

8.1 Mechanisms and Neural Substrates of One-Shot Integration

Consider a psychophysical experiment where an image of a real-world object has been severely degraded, thresholded, or heavily occluded to the point of appearing as entirely meaningless noise.[46] A naive observer might stare at this degraded image indefinitely without ever recognizing the hidden object.[46] However, if the observer is shown the clear, unoccluded, fully resolved version of the image for merely a fraction of a second, and then immediately returned to the degraded image, the hidden object instantly and vividly emerges from the noise.[46] This profound perceptual transformation is instantaneous, highly durable, and uniquely specific to that image.[15]

This phenomenon fundamentally relies on the brain's ability to instantly instantiate massive top-down priors to force perceptual completion across the visual array.[45] Once

the high-level visual cortex registers the true identity of the clear object, it immediately generates an overarching generative model.[42] Upon re-exposure to the degraded stimulus, this new model immediately cascades powerful predictive feedback down to the early visual cortex, guiding the spatial integration of the fragmented data points into a coherent Gestalt.[42]

Advanced neuroimaging utilizing ultra-high-field 7 Tesla (7T) fMRI and direct intracranial electroencephalography (iEEG) recordings have localized the neural substrates for these instantaneous priors primarily to the High-Level Visual Cortex (HLVC).[4342] Intracranial EEG reveals that prior-induced neural activity onsets first in the HLVC at approximately 225 milliseconds, directly preceding activity in the early visual cortex and confirming its role as the storage site for these top-down predictive templates.[44] To map the exact invariance properties of these stored priors, researchers have systematically manipulated the target images following the initial one-shot exposure.[46]

If the target image is scaled in size (altering the spatial frequencies processed by the small receptive fields of V1), the one-shot learning effect persists flawlessly.[46] However, if the image is spatially translated across the visual field by 6 degrees of visual angle to the left or right, or rotated/inverted, the perceptual completion effect significantly diminishes, though it is not entirely abolished.[46] This highly specific behavioral profile perfectly mirrors the physiological receptive field properties of the IT cortex, where translation and rotation invariance begin to emerge but remain tightly bound to the initial encoding geometry.[46] Interestingly, biasing image properties to selectively activate either the magnocellular or parvocellular visual pathways does not diminish the learning effect, suggesting that either pathway alone possesses the functional bandwidth to encode and deploy these powerful perceptual priors.[43]

Beyond the visual cortex, rapid pattern completion is fundamentally analogous to episodic memory formation in the hippocampus, where the CA3 subregion utilizes hetero-association to achieve online, one-shot storage of pattern sequences without the need for extended consolidation.

8.2 Implications for Artificial Intelligence

These profound biological findings have sparked intense interest and parallel development within the field of artificial intelligence.[43] Traditional Convolutional Neural Networks (CNNs) notoriously struggle with visual closure and amodal completion.[12]

When subjected to systematic tests utilizing Kanizsa triangles and heavily obscured objects, most CNNs fail to perceive the illusory contours or perform line completion, revealing a fundamental divergence from human visual intelligence.[12] Comprehensive analyses reveal that only specific architectures, notably VGG16 and DenseNet-121, consistently exhibit the Closure effect, while others show highly variable or negative results.[12]

By deeply studying the one-shot learning dynamics of the human IT cortex, computer scientists are currently attempting to embed top-down feedback loops into novel Vision Transformers.[42] By incorporating these biological feedback mechanisms, the resulting deep neural network models can accurately recapitulate and predict human behavioral performance on perceptual completion tasks, displaying prior knowledge highly similar to the actual neural codes mapped within the human HLVC.[42] The prior knowledge learnt and deployed by these advanced models is shown to be highly similar to the actual neural codes mapped within the human high-level visual cortex, bridging the gap between biological and artificial vision systems.[42]

9. Evolutionary Ecology and Adaptive Significance

The intricate neural machinery required to seamlessly fabricate missing visual data, propagate signals across hemifields, and maintain dynamic predictive models is remarkably metabolically expensive and computationally intensive.[39] From the rigorous perspective of evolutionary psychology and behavioral ecology, the persistence and universality of these completion mechanisms across species forcefully indicate a profound adaptive value tightly linked to survival and reproduction.[49]

9.1 The Survival Processing Advantage and Error Management

Evolutionary psychology posits that modern human cognitive architecture consists of specific psychological adaptations meticulously engineered by natural selection to solve recurrent challenges in ancestral hunter-gatherer environments.[49] Humans effectively navigate the modern world carrying "stone-age" brains, tuned to fitness-based problems.[51]

In the wild, predators and prey rarely stand in plain sight against highly contrasting, uniform backgrounds.[53] They are usually hidden behind dense foliage, rocky outcroppings, or varying terrain, creating massive visual occlusion. An organism whose visual system strictly limited its perception only to verified, bottom-up sensory input would perceive a leopard hidden in tall grass as merely a random, chaotic collection of

disconnected yellow and black patches. Such an organism would critically fail to rapidly recognize the object, fail to initiate a flight response, and consequently be rapidly selected out of the gene pool.[18]

The brain's uncompromising mandate to aggressively enforce the Law of Closure and synthesize amodal completion allows for split-second recognition of imminent threats from minimal visual cues.[18] This perfectly aligns with "Error Management Theory," a core evolutionary psychological framework describing the evolution of cognitive biases.[50] In an uncertain, high-risk environment, an organism faces two types of perceptual errors: a false positive (hallucinating a predator that isn't there) and a false negative (failing to synthesize the fragments of a predator that is genuinely there).[50] A false positive results in a minor metabolic cost—a brief surge of adrenaline and a wasted evasive maneuver. A false negative results in immediate death.[50] Consequently, the mammalian visual system has evolved a hyper-sensitive bias toward completion and holistic object formation, systematically erring on the side of synthesizing a complete, threatening object from fractional data rather than demanding absolute, lethal sensory confirmation.[18]

9.2 The Evolutionary Arms Race: Camouflage and Complexity

The robust capacity for amodal and modal completion in the visual systems of predators has, in turn, driven the spectacular evolutionary diversification of animal camouflage strategies.[53] Camouflage is fundamentally an ecological exploitation of the brain's Gestalt grouping principles, representing a visual arms race between the perceiver's interpolative algorithms and the prey's disruptive patterns.[53]

If an observer's visual system relies heavily on the Law of Good Continuation, relatability, and Closure to interpolate boundaries and identify targets, prey species will evolve body patterns designed explicitly to disrupt these very cues.[54] Disruptive coloration utilizes high-contrast internal markings that break up the continuous outline of the animal, creating false edges that force the predator's visual system to erroneously segment the prey into multiple, non-biological shapes.[54] This forces a failure in the amodal completion network, preventing the predator from synthesizing the fragments into a single, recognizable, selectable target.[54]

Simulations utilizing artificial neural networks modeling prey coloration in visually complex versus simple habitats confirm this evolutionary dynamic.[55] In highly complex visual backgrounds (such as dense rainforests or coral reefs), the sheer volume of fragmented edges creates massive sensory noise.[55] Prey animals that evolve coloration

matching this background complexity dramatically increase the computational difficulty of the detection task for the predator.[55] Selection on camouflage effectively exploits the inherent limitations and bottlenecks in the predator's information processing and contour completion networks.[55] Thus, the mechanisms of perceptual completion are not merely internal cognitive curiosities isolated in a laboratory; they are the active, driving frontlines in the evolutionary arms races that have shaped the morphological diversity, coloration, and behavior of the entire animal kingdom.[53]

10. Conclusion

The phenomenon of perceptual completion bridges the gap between raw sensory data and conscious phenomenological experience.[2] As demonstrated, the brain's ability to "create the other half" relies on an intricate, bidirectional cortical hierarchy. Feedforward structural heuristics (Gestalt principles) generated in the early visual cortex interface dynamically with massive top-down predictive priors stored in the high-level visual cortex. The discovery of one-shot perceptual learning further underscores the profound cognitive penetrability of vision, revealing a system optimized for rapid, survival-based inference rather than passive, objective recording. As artificial intelligence systems (such as advanced Vision Transformers) increasingly adopt these biologically inspired feedback loops [42], the synthesis of cognitive neuroscience and machine learning promises to fully decode the predictive algorithms that construct human visual reality.[45]

11. Declarations

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- **Conflicts of Interest:** Just about the thought that popped into my mind, and so I researched it.

Works cited

1. Neuroimaging Findings on Amodal Completion: A Review - PMC, https://pmc.ncbi.nlm.nih.gov/articles/PMC6457032/
2. Amodal completion across the brain: The impact of structure and knowledge | JOV, https://jov.arvojournals.org/article.aspx?articleid=2793751
3. Predictive Coding: A Possible Explanation of Filling-In at the Blind ..., https://pmc.ncbi.nlm.nih.gov/articles/PMC4784844/
4. Filling-in - Wikipedia, https://en.wikipedia.org/wiki/Filling-in

5. Amodal completion - Wikipedia, https://en.wikipedia.org/wiki/Amodal_completion
6. The Importance of Amodal Completion in Everyday Perception - PMC, https://pmc.ncbi.nlm.nih.gov/articles/PMC6083800/
7. Development of modal and amodal completion in infants - PubMed, https://pubmed.ncbi.nlm.nih.gov/17120844/
8. What, if anything, can be considered an amodal sensory dimension? - PMC, https://pmc.ncbi.nlm.nih.gov/articles/PMC11543734/
9. A new taxonomy for perceptual filling-in - PMC - NIH, https://pmc.ncbi.nlm.nih.gov/articles/PMC3119792/
10. Illusory contours: a window onto the neurophysiology of constructing perception, https://cibm.ch/wp-content/uploads/1-s2.0-S1364661313001502-main.pdf
11. Why the strange triangle?, https://www.princeton.edu/~freshman/kanizsa.html
12. (PDF) Finding Closure: A Closer Look at the Gestalt Law of Closure in Convolutional Neural Networks - ResearchGate, https://www.researchgate.net/publication/383308594_Finding_Closure_A_Closer_Look_at_the_Gestalt_Law_of_Closure_in_Convolutional_Neural_Networks
13. A Closer Look at the Gestalt Law of Closure in Convolutional Neural Networks - arXiv, https://arxiv.org/html/2408.12460v1
14. The Representation of Illusory and Real Contours in Human Cortical Visual Areas Revealed by Functional Magnetic Resonance Imaging | Journal of Neuroscience, https://www.jneurosci.org/content/19/19/8560
15. Neuronal signatures of successful one-shot memory in mid-level visual cortex - PMC, https://pmc.ncbi.nlm.nih.gov/articles/PMC12485755/
16. Gestalt psychology - Wikipedia, https://en.wikipedia.org/wiki/Gestalt_psychology
17. A Century of Gestalt Psychology in Visual Perception I. Perceptual ..., https://pmc.ncbi.nlm.nih.gov/articles/PMC3482144/
18. The Law of Closure Shapes Perception: The Psychology Behind It - Helio, https://helio.app/ux-research/laws-of-ux/law-of-closure/
19. Amodal completion in visual perception, https://www.mi.sanu.ac.rs/vismath/fila/
20. Perceptual organization and visual awareness: the case of amodal completion - Frontiers, https://www.frontiersin.org/journals/psychology/articles/10.3389/fpsyg.2023.1201681/full
21. Influence of Complexity and Gestalt Principles on Aesthetic Preferences for Building Façades: An Eye Tracking Study - PMC, https://pmc.ncbi.nlm.nih.gov/articles/PMC11379339/
22. The effects of Kanizsa contours on temporal integration and attention in rapid serial visual presentation - PMC, https://pmc.ncbi.nlm.nih.gov/articles/PMC5516004/
23. When Do Visual Category Representations Emerge in Infants' Brains? - eLife, https://elifesciences.org/reviewed-preprints/100260v1/pdf
24. How Infants Learn About the Visual World - UCLA Baby Lab,

https://www.babylab.ucla.edu/wp-content/uploads/sites/8/2016/09/Johnson2010.pdf
25. Perception of illusory contours in children and adults: An eye-tracking study - PMC, https://pmc.ncbi.nlm.nih.gov/articles/PMC11480146/
26. Image parsing mechanisms of the visual cortex - Johns Hopkins Computer Science, https://www.cs.jhu.edu/~ayuille1/courses/Stat271-Fall13/VonderHeydt_ImageParsing_neuroscience.pdf
27. The neural mechanisms of perceptual filling-in - ResearchGate, https://www.researchgate.net/publication/7281347_The_neural_mechanisms_of_perceptual_filling-in
28. McGruer, Fiona (2017) Basic prediction mechanisms as a precursor for schizophrenia studies. PhD thesis. - Enlighten Theses, https://theses.gla.ac.uk/8236/13/2017McGruerPhD.pdf
29. Amodal completion instead of predictive coding can explain activity suppression of early visual cortex during illusory shape perception - Journal of Vision, https://jov.arvojournals.org/article.aspx?articleid=2772601
30. Amodal completion and relationalism, https://d-nb.info/1262143160/34
31. Perceptual Completion across the Vertical Meridian and the Role of Early Visual Cortex - pillow lab @ princeton, https://pillowlab.princeton.edu/pubs/Pillow_Rubin_02.pdf
32. Perceptual completion across the vertical meridian and the role of early visual cortex - PubMed, https://pubmed.ncbi.nlm.nih.gov/11879656/
33. Neural Mechanisms of Spatial Selective Attention in Areas V1, V2, and V4 of Macaque Visual Cortex | Journal of Neurophysiology | American Physiological Society, https://journals.physiology.org/doi/10.1152/jn.1997.77.1.24
34. Finding out about filling-in: A guide to perceptual completion for visual science and the philosophy of perception - Royal Holloway Research Portal, https://pure.royalholloway.ac.uk/ws/files/4739153/S0140525X9800175Xa.pdf
35. Quasi-Modal Encounters Of The Third Kind: The Filling-In Of Visual Detail - Swarthmore College, https://works.swarthmore.edu/cgi/viewcontent.cgi?referer=&httpsredir=1&article=1032&context=fac-psychology
36. Amodal completion instead of predictive coding can explain activity suppression of early visual cortex during illusory shape perception - PMC, https://pmc.ncbi.nlm.nih.gov/articles/PMC8131992/
37. Amodal completion and relationalism - PMC - NIH, https://pmc.ncbi.nlm.nih.gov/articles/PMC9287258/
38. Predictive coding - Wikipedia, https://en.wikipedia.org/wiki/Predictive_coding
39. Computational components of visual predictive coding circuitry - Frontiers, https://www.frontiersin.org/journals/neural-circuits/articles/10.3389/fncir.2023.1254009/full
40. Amodal completion instead of predictive coding can explain activity suppression of

early visual cortex during illusory shape perception - ResearchGate, https://www.researchgate.net/publication/351597456_Amodal_completion_instead_of_predictive_coding_can_explain_activity_suppression_of_early_visual_cortex_during_illusory_shape_perception
41. Top Down Effects of Cognition on Perception, https://perception.yale.edu/Brian/refGuides/TopDown.html
42. Neural and computational mechanisms underlying one-shot perceptual learning in humans, https://pubmed.ncbi.nlm.nih.gov/41639076/
43. Neural and computational mechanisms underlying one-shot perceptual learning in humans - PMC, https://pmc.ncbi.nlm.nih.gov/articles/PMC12873369/
44. Neural and Computational Mechanisms Underlying One-shot Perceptual Learning in Humans | bioRxiv, https://www.biorxiv.org/content/10.1101/2025.05.09.635727v1.full-text
45. Neural and Computational Mechanisms Underlying One-shot Perceptual Learning in Humans | bioRxiv, https://www.biorxiv.org/content/10.1101/2025.05.09.635727v1
46. Mapping the invariance properties of perceptual priors in one-shot perceptual learning | JOV, https://jov.arvojournals.org/article.aspx?articleid=2792592
47. Study identifies brain region driving one-shot visual learning - News-Medical.Net, https://www.news-medical.net/news/20260204/Study-identifies-brain-region-driving-one-shot-visual-learning.aspx
48. Neuronal signatures of successful one-shot memory in mid-level visual cortex - bioRxiv, https://www.biorxiv.org/content/10.1101/2025.09.22.677855v1.full.pdf
49. Evolutionary psychology - Wikipedia, https://en.wikipedia.org/wiki/Evolutionary_psychology
50. Evolutionary Theories in Psychology - Noba Project, https://nobaproject.com/modules/evolutionary-theories-in-psychology
51. Evolutionary Constraints on Remembering - Adaptive Memory Lab, http://evo.psych.purdue.edu/downloads/2010_Nairne_Evolutionary_Constraints_on_Remembering.pdf
52. Adaptive memory: ancestral priorities and the mnemonic value of survival processing - PubMed, https://pubmed.ncbi.nlm.nih.gov/20206924/
53. Camouflage - Exeter Visual Ecology, https://www.visual-ecology.com/camouflage/
54. Camouflage and visual perception - PMC - NIH, https://pmc.ncbi.nlm.nih.gov/articles/PMC2674079/
55. Visual Background Complexity Facilitates the Evolution of Camouflage - PubMed, https://pubmed.ncbi.nlm.nih.gov/12894933/
56. Optimizing colour for camouflage and visibility using deep learning: the effects of the environment and the observer's visual system - Royal Society Publishing, https://royalsocietypublishing.org/rsif/article/16/154/20190183/87081/Optimizing-colour-for-camouflage-and-visibility